

Database Management Systems (DBMS) Notes

A Database Management System (DBMS) is software that manages databases, allowing users to create, read, update, and delete data efficiently. It ensures data integrity, security, and supports concurrent access by multiple users.

Normalization

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. The main normal forms are First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF), and Boyce-Codd Normal Form (BCNF). Each normal form has specific rules: 1NF requires atomic values in each column, 2NF removes partial dependency on a composite key, and 3NF removes transitive dependency, ensuring non-key attributes depend only on the primary key.

SQL Joins

Joins are used in SQL to combine rows from two or more tables based on a related column. The main types of joins are INNER JOIN, which returns records with matching values in both tables; LEFT JOIN, which returns all records from the left table and matched records from the right; RIGHT JOIN, which does the opposite; and FULL OUTER JOIN, which returns all records when there is a match in either table.

ACID Properties

ACID stands for Atomicity, Consistency, Isolation, and Durability, which are four key properties that guarantee reliable processing of database transactions. Atomicity ensures a transaction is all-or-nothing. Consistency ensures the database remains in a valid state before and after the transaction. Isolation ensures concurrent transactions do not interfere with each other. Durability ensures that once a transaction is committed, it remains so even in the event of a system failure.

Indexing

An index in a database is a data structure that improves the speed of data retrieval operations at the cost of additional storage space and slower write operations. Common indexing structures include B-Trees and Hash Indexes. Indexes are typically created on columns that are frequently used in WHERE clauses or as JOIN conditions to speed up query performance.